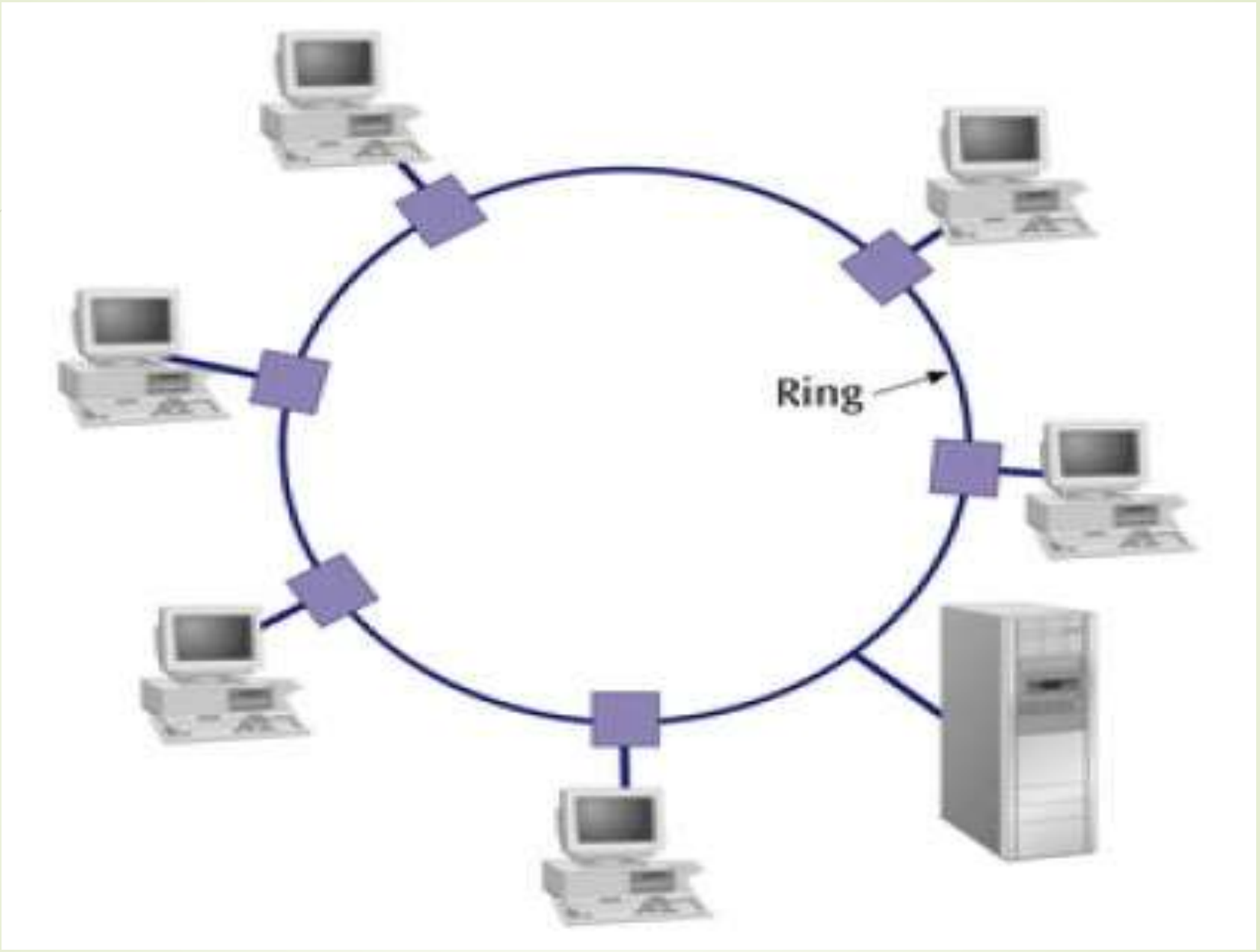
Ring topology :-

- A ring network is one where all workstations and other devices are connected in a continuous loop. There is no central server
- started out as a simple peer-to-peer LAN topology.
- Each networked workstation had two connections: one to each of its nearest neighbors
- Data was transmitted unidirectionally around the ring
- Sending and receiving of data takes place by the help of TOKEN

Token Passing ;-

- Token contains a piece of information which along with data is sent by the source computer
- This token then passes to next node, which checks if the signal is intended to it


Ring

Advantages of Ring topology :-

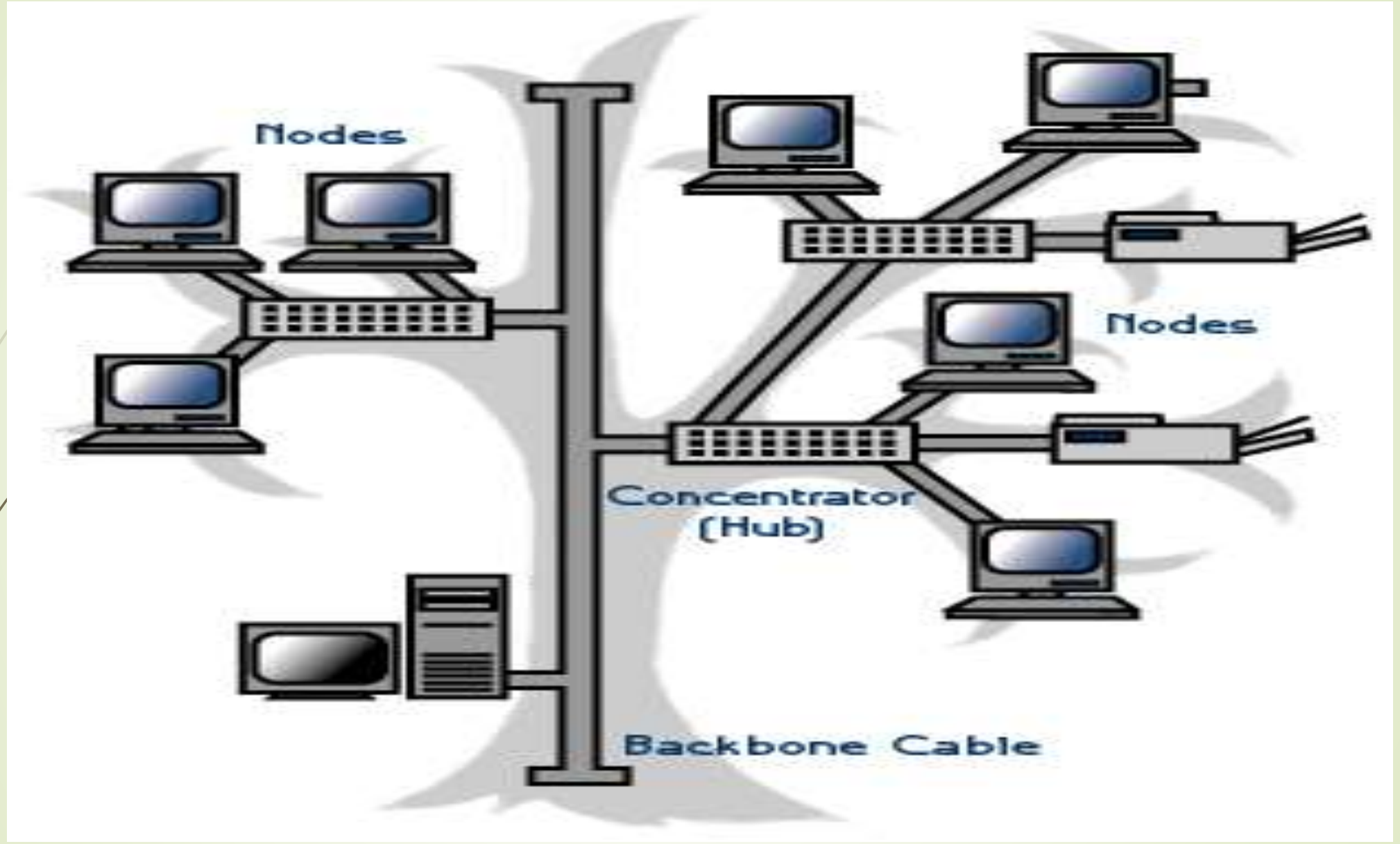
1) This type of network topology is very organized
2) Performance is better than that of Bus topology
3) Does not require a central node to manage the connectivity between the computer.
4) Additional components do not affect the performance of network
5) Each computer has equal access to resources

Disadvantages of Ring topology :-

1) Each packet of data must pass through all the computers between source and destination, slower than star topology
2) If one workstation or port goes down, the entire network gets affected
3) Network is highly dependent on the wire which connects different components

Tree topology :-

- Tree Topology integrates (link / add)the characteristics of Star and Bus Topology.
- computers (nodes) are connected by each other through central hub. And Bus Topology, work station devices are connected by the common cable called Bus.
- After understanding these two network configurations, we can understand tree topology better.
- In Tree Topology, the number of Star networks is connected using Bus.


Nodes
Nodes
Concentrator
(Hub)
Backbone Cable

Advantages of Tree Topology :-

- 1. It is an extension of Star and bus Topologies.
- 2. Expansion (increase) of Network is possible and easy.

3. Here, we divide the whole network into segments (star networks), which can be easily managed and maintained.

4. Error detection and correction is easy.

5. Each segment is provided with dedicated point-to-point wiring to the central hub.

6. If one segment is damaged, other segments are not affected.

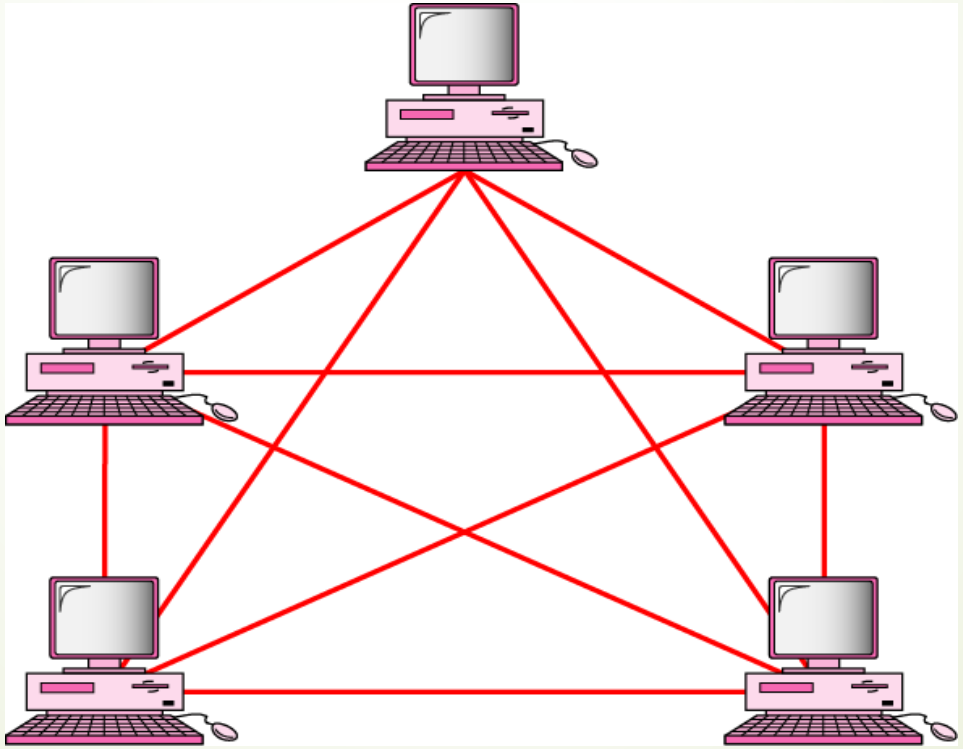
Disadvantages of Tree Topology:-

- 1. Because of its basic structure, tree topology, relies heavily on the main bus cable, if it breaks whole network is corrupted.
- 2. As more and more nodes and segments are added, the maintenance becomes difficult.
- 3. Scalability of the network depends on the type of cable used.

Mesh Topology:-

- • Each computer connects to every other.
- It is a point to point connection to other nodes or devices.
- All the network nodes are connected to each other.
- – Wiring is very complicated.
- – Cabling cost is high
- – Troubleshooting a failed.(cable)
- All computers connected together.
- Internet is a mesh network.

- – A variation hybrid mesh – create point to point connection between specific network devices, often seen in WAN implementation.



Mesh Advantages:-

- Communication is very fast between two nodes.
- The network can be expanded without disruption to current use.
- Data will always be delivered.
- A failure of one device does not cause a break in the network or transmission of data.
- Adding additional devices does not disrupt(break) data transmission between other devices.
- Provide security and privacy

Mesh Disadvantages:-

- Requires more cable than the other LAN topologies.
- Complicated implementation.
  - Lots of cable
  - Installation and configuration is difficult.**(Hard to setup)**
- It is the more expensive network from the point of view of link cost.